STAT3006 Worksheet 03

1. Likelihood-Based Loss Functions

Let $p(y|\mathbf{x};\boldsymbol{\theta})$ be a parametric model for the conditional distribution of y given $\mathbf{x}$.

(a) Define the negative log-likelihood (NLL) loss $\ell(y,\mathbf{x},\boldsymbol{\theta})$

Answer:

$$\ell(y,\mathbf{x},\boldsymbol{\theta}) = -\log\left\{p(y|\mathbf{x},\boldsymbol{\theta})\right\}$$

(b) Given i.i.d data $\{(\mathbf{x}_i, y_i)\}_{i=1}^{n}$, write down the emperical risk (average loss) $L(\boldsymbol{\theta})$

Answer:

$$L(\boldsymbol{\theta}) = \frac{1}{n}\sum_{i=1}^{n} -\log\left\{p(y_i|\mathbf{x}_i;\boldsymbol{\theta})\right\}$$

(c) Explain briefly why specifying a likelihood model is equivalent to specifying a loss function

Answer: since the loss function directly involves the likelihood function, choosing the likelihood directly affects the behaviour of the loss function and ultimately affects the model training.

(d) Consider the heteroskedastic Guassian model

$$p(y|\mathbf{x},\boldsymbol{\theta}) = \mathcal{N}(f(\mathbf{x};\boldsymbol{\theta}), \exp(h(\mathbf{x},\boldsymbol{\theta})))$$

Dervie the negative log-likelihood loss (up to additive constants independent of $\boldsymbol{\theta}$

Answer: The distribution function of the above Guassian model is given by

$$p(y|\mathbf{x},\boldsymbol{\theta}) = (2\pi\exp(h(\mathbf{x};\boldsymbol{\theta})))^{-1/2}\exp\left\{\frac{-(\mathbf{x}-f(\mathbf{x},\boldsymbol{\theta}))^2}{2\exp(h(\mathbf{x},\boldsymbol{\theta}))}\right\}$$

Taking the negative log we get

$$\log p = \frac{-(\mathbf{x}-f(\mathbf{x},\boldsymbol{\theta}))^2}{2\exp(h(\mathbf{x},\boldsymbol{\theta}))} - \frac{1}{2}\log(2\pi) - \frac{1}{2}\log(\exp(h(\mathbf{x},\boldsymbol{\theta})))$$
$$-\log p = \frac{(\mathbf{x}-f(\mathbf{x},\boldsymbol{\theta}))^2}{2\exp(h(\mathbf{x},\boldsymbol{\theta}))} + \frac{1}{2}\log(2\pi) + \frac{1}{2}\log(h(\mathbf{x},\boldsymbol{\theta}))$$

(e) Interpret the loss from part (d). In particular, explain how the model treats prediction errors when $\exp(h(\mathbf{x},\boldsymbol{\theta}))$ is large versus small, and why the additional term $h(\mathbf{x},\boldsymbol{\theta})$ is necessary.

Answer: When $\exp(h(\mathbf{x}, \boldsymbol{\theta}))$ is large, the squared prediction error (first term) is discounted and the loss of an observation is attributed to the model's own uncertainty via the variance (third term).

When $\exp(h(\mathbf{x}, \boldsymbol{\theta}))$ is small, the loss is attributed more towards the squared prediction error, and the closer the prediction is to the true value the smaller the loss.

The $h(\mathbf{x}, \boldsymbol{\theta})$ term is necessary because without the third term, we can push the variance up to infinty and it would look like the model is performing very well as the loss is small. The model would actually never be able to learn to predict y

2. Bayes Optimal Predictor (Categorical Likelihood)

Let $y \in \{1, 2, \ldots, K\}$ and suppose a predictor

$$\mathbf{f} : \mathcal{X} \to \Delta^{K-1}$$

outputs a probability vector

$$\mathbf{f}(\mathbf{x}) = \Big(f_1(\mathbf{x}), \ldots, f_K(\mathbf{x})\Big), \quad f_k(\mathbf{x}) \geq 0, \quad \textstyle\sum_{k=1}^{K} f_k(\mathbf{x}) = 1$$

Here, Δ^{K-1} denotes the probability simplex:

$$\Delta^{K-1} = \left\{ \mathbf{t} \in \mathbb{R}^K : t_k \geq 0, \sum_{k=1}^{K} t_k = 1 \right\}$$

Consider the negative log-likelihood (cross-entropy) loss:

$$\ell(y, \mathbf{f}(\mathbf{x})) = -\sum_{k=1}^{K} \mathbf{1}\{y = k\} \log f_k(\mathbf{x})$$

For a fixed input $\mathbf{x}$, find the Bayes optimal predictor

$$\mathbf{f}^*(\mathbf{x}) = \underset{\mathbf{t} \in \Delta^{K-1}}{\arg\min}\, \mathbb{E}_{p^*(y|\mathbf{x})}[\ell(y, \mathbf{t})]$$

Answer: First we find the conditional risk of the loss function, $\mathbb{E}_{p^*(y|\mathbf{x})}[\ell(y, \mathbf{t})]$,

$$
\begin{aligned}
\mathbb{E}_{p^*(y|\mathbf{x})}[\ell(y, \mathbf{t})] &= \mathbb{E}\left\{ -\sum_{k=1}^{K} \mathbf{1}\{y = k\} \log t_k \right\} \\
&= -\sum_{k=1}^{K} p_k^* \log t_k
\end{aligned}
$$

We can now use the Lagrange Multiplier Method and form the Langrangian

$$
\mathcal{L}(\mathbf{t}, \lambda) = -\sum_{k=1}^{K} p_k^* \log t_k + \lambda \left(\sum_{k=1}^{K} t_k - 1 \right)
$$

Taking the derivative with respect to t_k we get

$$
\frac{\partial \mathcal{L}}{\partial t_k} = -\frac{p_k^*}{t_k} + \lambda
$$

Letting the L.H.S. equal to zero we get

$$
t_k = \frac{p_k^*}{\lambda}
$$

Subbing this value of t_k into our constraint we can solve for λ,

$$
\begin{aligned}
\sum_{k=1}^{K} \frac{p_k^*}{\lambda} &= 1 \\
\frac{1}{\lambda}(1) &= 1 \\
\lambda &= 1
\end{aligned}
$$

Subbing this value back into t_k we get

$$
t_k = p_k^*
$$

Therefore the Bayes optimal predictor for the categorical cross-entropy loss is exactly the true conditional distribution